



[SRS Home](#) | [Front Page](#) | [Monthly Issue](#) | [Index](#)



☐ Search WWW ☒ Search seattlerobotics.org

The Basics

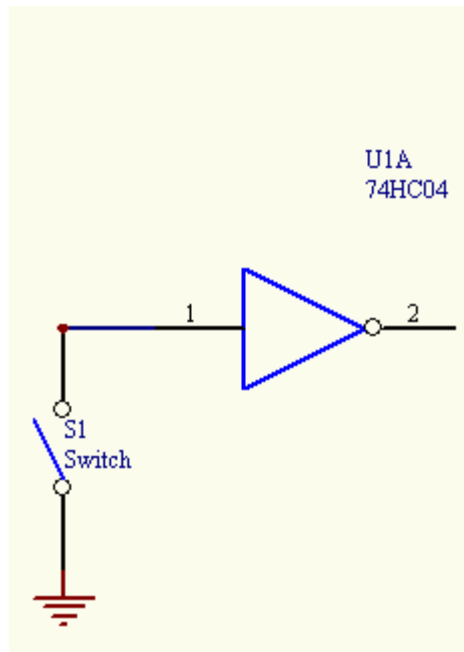
This column is intended to be a place to find really basic information on subjects related to robotics and electronics. If you have questions about the basics on these subjects, this column is for you. I invite you to submit questions or information to this column. Send me some mail.

[Kevin Ross](#)

Very Basic Circuits

The Pull-up Resistor

This time in Very Basic Circuits, I would like to talk about pull-up resistors. The basic function of a pull-up resistor is to insure that given no other input, a circuit assumes a default value. Actually, there are two flavors of this circuit. A pull-up and a pull-down. Their function is the same, to create a default value for a circuit, but one pulls the line high, the other pulls it low.



Consider this schematic. The gate U1A has an input (pin 1) and an output (pin 2). The input state of most logic gates is called a high impedance. This means it provides no real power of its own. Therefore, if nothing is connected to pin 1, the value of the input is considered to be floating. Most gates will float towards a high state. This is a very weak condition, and any electrical noise could cause the input to go low.

When switch S1 is closed (on), the input state at pin1 goes low. Since there is a definite connection to an electrical potential (in this case ground), the state of the pin is stable.

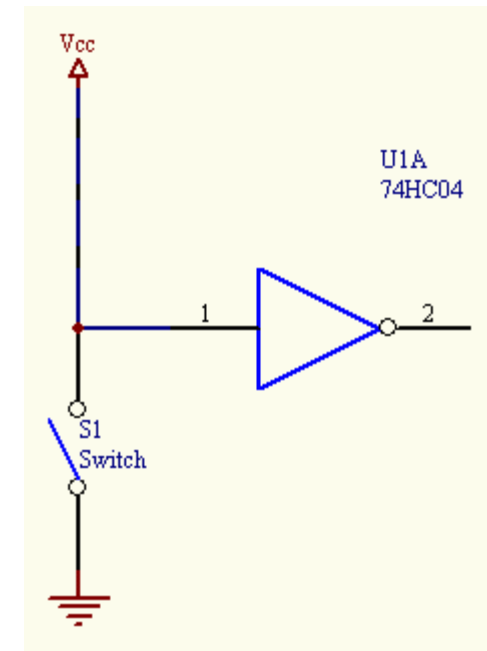
When switch S1 is open (off), then input pin 1 is susceptible to a wide array of electrical problems. The traces or wires connected to pin 1 may very well allow enough electrical noise in (by acting as little antennas) to cause pin 1 to incorrectly switch states. What is needed here is a way to connect pin 1 to an electrical potential that can be removed when the switch is closed. This electrical potential will allow the pin to keep a steady state.

A floating input gate. Not Good!

One thought is to tie the pin to Vcc (+5 volts) to insure that pin 1 doesn't float. The circuit to the right certainly does that. With pin 1 tied directly to Vcc, the line does not float, and has an ON state.

The problem with this circuit is what happens when switch S1 is closed. This creates a direct electrical connection between Vcc and GND. In other words, it will short out the circuit. If you are lucky, it will just stop your entire system from working. If you are unlucky, it will burn up the wires!

The problem with short circuits is they allow too much current to flow from Vcc to GND. This causes heat to be generated, which can sometimes burn parts, wires, or even start fires. In addition, most circuits fail to function correctly because the voltage at the power supply drops to zero. In general, this is a bad situation!



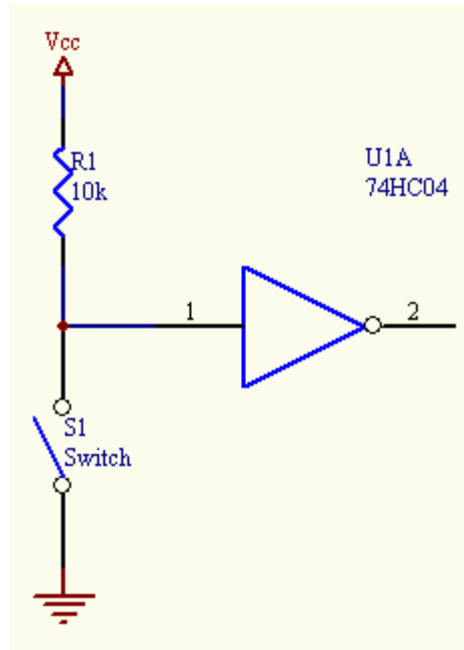
A very bad idea!

Now consider the next schematic, which is similar to the first but has added a pull-up resistor. This resistor's function is to limit the amount of current that can flow through the circuit.

When switch S1 is open (off), pin 1 is tied to Vcc through the resistor. Since pin 1 is a high impedance input, a voltage meter or logic probe placed on pin 1 will show Vcc (+5v) if connected to pin 1.

When switch S1 is closed (on), pin 1 has a direct connection to GND, which takes it to the low state. The pin 1 side of R1 also has a direct connection to ground. Current will flow from Vcc, through R1, and to ground. It isn't considered a short, however, because R1 will limit the amount of current that can flow to a very small amount. In fact, you can compute this using Ohm's law.

$$I = V / R$$



$$I = 5\text{V} / 10,000\text{ohms}$$

$$I = .0005\text{A} (.5\text{mA})$$

Pull-up resistor limits the current

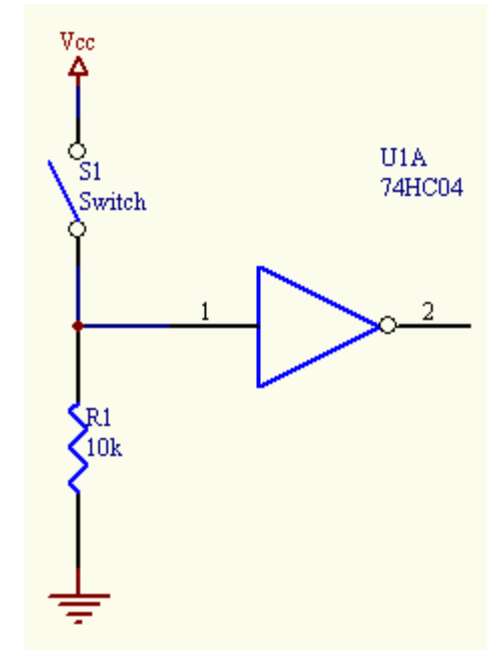
A variation on this them is a pull-down resistor. Just like the pull-up resistor, it is used to limit the current that can flow between Vcc and ground. Though less often used, it is still a valid thing to do.

Most digital circuits use a 10k or a 47k resistor for pullups. The exact value doesn't actually matter, as long as it is high enough to prevent too much current from flowing. 10k seems to be the most common, but if you are hoping to save as much power as possible, the a 47k resistor may be right for your application. In some cases, you can go higher, but then you are depending on characteristics of the pins on the chip.

In Summary

You will find that pull-up resistors are extremely common is most digital circuits. The key function

for a pull-up is to prevent input lines from floating. The key function for the resistor itself is to prevent too much current from flowing through the pull-up circuit.



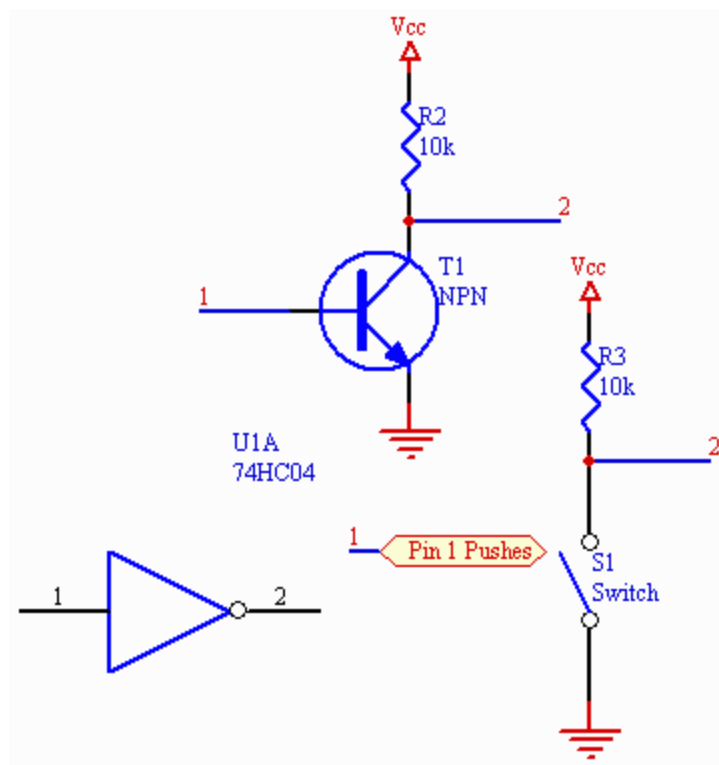
The less common pull-down

Very Basic Circuits

The Current Limiting Resistor

If you have just read the above article, you hopefully understand how a resistor limits the amount of current that can flow through a circuit. In the above example, we were dealing with input pins. Expanding on that idea, I would like to present a simple circuit that use limiting resistors with output pins and also with an LED.

Unlike an input pin, which has only high impedance, an output pin is designed to



have two states: Drive (on, or high logic) and Sink (off, or low logic).

Let us look at a simple digital circuit, the 74HC04 Inverter. The picture on the right shows the symbol for the 74HC04. Internally, this chip is constructed using transistors. I have taken great liberty by reducing the circuit to a single transistor version, T1. Digital circuits use transistors as switches. When a current is supplied at pin 1, the transistor allows current to flow from pin 2 to GND.

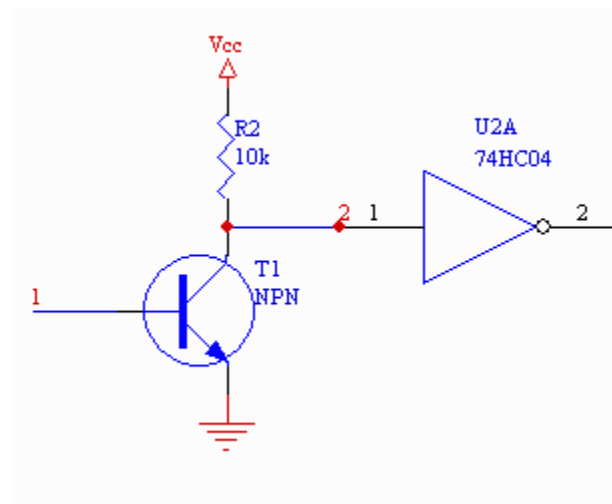
I have included a picture of a switch as well. Closing the switch allows current to flow from pin 2 to GND. These are functionally equivalent circuits. The actual implementation of the 74HC04 is much more complex, but the basic ideas presented are still valid.

In digital circuits, output gates are switches.

In the above drawing, you can see that the equivalent circuit using the switch demonstrates that the internals of an output gate use a pull-up resistor just like I described in the first article. By replacing the original push button switch by the equivalent transistor circuit, it looks like the schematic on the right.

A quick analysis shows that when T1 is off, R2 pulls the output pin high, which is good for U2A's input pin. When T1 is on, T1 ties U2A's input pin to GND, which brings the input pin low. R2 allows a little bit of current to flow.

Note that on the real 74HC04, the pull-up resistor R2 is internal to the chip. Therefore, you can directly connect the output of most integrated circuits directly to the input of others without an external pull-up resistor.

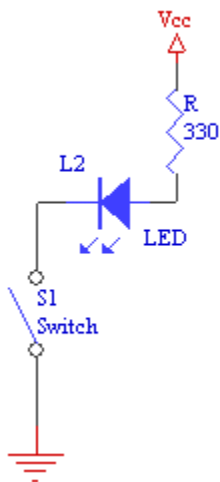


There are times when you connect the output of a gate to a device that isn't the input of another device. For example, driving an LED, is an example.

An important consideration when connecting anything to an output gate is what will happen when the gate drops to logic zero. This basically creates a direct line to GND, just like the switch example. Before you connect anything to an output gate, you should consider how much current will flow through the gate in the zero state. Quite often, you will need to add a current limiting resistor to insure that you don't burn up the part.

Most logic parts are capable of handling around 20mA of current per pin. This means that you need to consider carefully the device being attached.

How to make an LED work



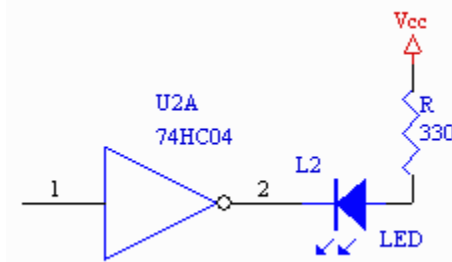
An LED (Light Emitting Diode) is a semiconductor that emits light energy when a current flows through it. Current will only flow one direction, just like a regular diode. There are a few things you need to know about an LED before you use one. First, and most importantly, is that an LED has very low internal resistance. This means that left to itself, an LED will pass so much current that it will burn up. They require an external resistor to limit the current.

Most LED's have a current rating, which determines the size of the resistor you will need. The current rating tells you what the maximum allowable current for the part is. In general, the higher the current, the brighter the LED.

Most LED's seem to handle at least 15mA. If you are using a 5 volt circuit, then Ohms law tells you what resistor value to use. $R = V / I$, so $R = 5v / .015A = 333 \text{ ohms}$.

Now let us consider what happens when using the output of a chip, such as the 74HC04, to operate an external device. For example, the circuit on the right drives an LED. When the gate is HIGH, then there is no path to GND for cathode of the LED L1. When the gate is LOW, then output pin 2 is connected to ground, and current flows.

Since R3 only allows 15mA of current to pass, the gate is safe from being overloaded. Remember that most gates can handle 20mA of current. The same holds true for most microcontrollers.



In Summary

The output from gates acts very much like a switch. When the gate is logic low, it goes to ground. Most gates can only handle about 20mA of current without burning up. You should always understand how much current will flow when the device is connected to a logic low gate.

What else do you want to know? Send me mail and ask away. [Kevin Ross](#)